Computer Vision

Homework 2: Hand Shapes and Gestures

Hieu Nguyen

Based on Assignment 2 of CS 585 (Spring 2024), a programming assignment on hand interaction. Teams of two are fine. Goes with Chapter 1 of the notes.

Build a program that watches a webcam, recognizes hand shapes or gestures, and drives a graphical display that reacts to them.

1. **Recognition.** Recognize at least four different hand shapes or gestures, for example a fist, thumbs up or down, a pointing index finger, waving, or drawing a shape in the air.
2. **Methods.** Use at least two of the methods from Chapter 1, with an emphasis on the analysis of binary shapes: projections and bounding boxes, background or frame differencing, motion energy and motion history images, skin-colour thresholding, template matching with normalized correlation, the centroid, orientation and roundness of a blob, and simple tracking. You may call OpenCV functions, but you should be able to explain the mathematics of every one you use on a whiteboard.
3. **Graphics.** Make a graphical display that responds to the recognized shapes in a way that is tasteful and fun to use.
4. **Evaluation.** Collect your own test videos (different people, lighting and backgrounds if you can) and report how well the recognizer works with a confusion matrix, as in Section 4.2.2. Discuss the most common confusions and why they happen.

Hand in your code, a short report (PDF) with the confusion matrix and a description of your methods and of how the graphics respond to each gesture, and a short video of the system running live.